

IF Function

Logical

	A	B	C	D	E	F
1	IF					
2						
3	Student Roll	Student Name	Gender	Marks	Status	
4	1	Arif Hossain	Male	97	Pass	
5	2	Jihad Ahmed	Male	23	Fail	
6	3	Farhan Shikder	Male	97	Pass	
7	4	Khadija Akther	Female	33	Pass	
8	5	Rahul Mia	Male	71	Pass	
9	6	Raiyan	Male	32	Fail	
10	7	Shneha Akther	Female	75	Pass	
11	8	Karim Uddin	Male	50	Pass	
12	9	Arafat Hossain	Male	47	Pass	
13	10	Saiful Islam	Male	23	Fail	
14						
15						
16	Student Roll	Student Name	Gender	Name MD/Miss		
17	1	Arif Hossain	Male	MD Arif Hossain		
18	2	Jihad Ahmed	Male	MD Jihad Ahmed		
19	3	Farhan Shikder	Male	MD Farhan Shikder		
20	4	Khadija Akther	Female	Miss Khadija Akther		
21	5	Rahul Mia	Male	MD Rahul Mia		
22	6	Raiyan	Male	MD Raiyan		
23	7	Shneha Akther	Female	Miss Shneha Akther		
24	8	Karim Uddin	Male	MD Karim Uddin		
25	9	Arafat Hossain	Male	MD Arafat Hossain		
26	10	Saiful Islam	Male	MD Saiful Islam		
27						

IFS Function

Logical

	A	B	C	D
1	IFS			
2				
3		Age	Age Group	
4		35	30-39	
5		39	30-39	
6		33	30-39	
7		20	20-29	
8		47	40-49	
9		64	60-69	
10		58	50-59	
11		62	60-69	
12		67	60-69	
13		51	50-59	
14		24	20-29	
15		61	60-69	
16		39	30-39	
17		31	30-39	
18		37	30-39	
19		20	20-29	
20		70	Senior Citizen	
21		26	20-29	
22		46	40-49	
23		23	20-29	
24		55	50-59	
25		87	Senior Citizen	
26		39	30-39	
27		76	Senior Citizen	
28		58	50-59	
29		61	60-69	
30				
31				

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
4	Student Roll	Student Name	Gender	Ban	Eng	Math	Status			IFS					
5	1	Arif Hossain	Male	97	51	97	Pass								
6	2	Jihad Ahmed	Male	23	15	29	Fail								
7	3	Farhan Shikder	Male	97	74	30	Fail								
8	4	Khadija Akther	Female	33	90	99	Pass								
9	5	Rahul Mia	Male	71	79	80	Pass								
10	6	Raiyan	Male	23	95	35	Fail								
11	7	Shneha Akther	Female	75	80	100	Pass								
12	8	Karim Uddin	Male	50	97	82	Pass								
13	9	Arafat Hossain	Male	47	80	71	Pass								
14	10	Saiful Islam	Male	23	67	31	Fail								
15															
16										Bonus >	Bonus %				
17										85000	5%				
18															
19															
20															
21	Salary	EMP Name	Week 1	Week 2	Week 3	Week 4	Total Sales	Bonus Status	Bonus Amount	Bonus With salary					
22	20000	Arafat Hossain	24382	15000	24382	18000	81764	No Bonus	0	20000					
23	35000	Sajid Khan	24382	24382	10000	24382	83146	No Bonus	0	35000					
24	40000	Rifat Hossain	23088	25110	21498	15998	85694	Get Bonus	4284.7	44284.7					
25	25000	Sagor Khan	20725	25321	19312	28131	93489	Get Bonus	4674.45	29674.45					
26	27000	Raiyan Ahmed	15762	17759	16690	27171	77382	No Bonus	0	27000					
27															
28							Total Sales		421475		Team Bonus				
29							Avg Sales		84295		Better Luck Next Time				
30							Max Sales		93489						
31							Min Sales		77382						

Total Sale >= \$ 400,000
AND
Avg Sale > \$ 85,000

Lookup

Lookup

[illegible]

XLOOKUP Function

Lookup

[illegible]

Lookup

Lookup

[illegible]

Functions at a Glance

LOGICAL FUNCTIONS

IF

```
=IF(test, value_if_true, value_if_false)
```

- Tests a single condition
- Returns one value if TRUE, another if FALSE
- Can be nested but gets complex quickly

IFS

```
=IFS(cond1, val1, cond2, val2, ...)
```

- Tests multiple conditions in sequence
- Returns value for first TRUE condition
- Cleaner alternative to nested IF

LOOKUP FUNCTIONS

VLOOKUP

```
=VLOOKUP(val, array, col, 0)
```

- Searches vertically (top to bottom)
- Can only look left-to-right

XLOOKUP

```
=XLOOKUP(val, lookup, return)
```

- Modern replacement for VLOOKUP
- Searches in any direction

HLOOKUP

```
=HLOOKUP(val, array, row, 0)
```

- Searches horizontally (left to right)
- Returns value from a specified row

INDEX+MATCH

```
=INDEX(range, MATCH(val, col, 0))
```

- Most flexible lookup combo
- Works in any direction, any column